

Performance Evaluation and Design Method of Lifting-Wing Multicopters

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Abstract—As a novel vertical takeoff and landing unmanned aerial vehicle, a lifting-wing multicopter with simple structure, good stability, and longer flight range has been unveiled to be in service. However, how to make range as long as possible to realize its potential advantages is one of the key issues in the practical application of the lifting-wing multicopter. Thus, this study aims to propose a range and transition time performance evaluation to realize a sizing optimization design method for the lifting-wing multicopter. First, the dynamic model and the propulsion system model are established. Based on these models, evaluation methods for range and transition time performance are obtained with the consideration of mounting angle. Second, the influences of the main parameters on range performance are studied to offer references for design. Then, an explicit design process is proposed. Finally, a prototype is implemented, and flight experiments are performed to verify the evaluation and design method. The results indicate that the performance evaluation method has enough accuracy and suggest that it is possible to design a lifting-wing multicopter using the presented design method.

Index Terms—Aircraft, aircraft propulsion, design optimization, propellers, unmanned aerial vehicles (UAV).

I. INTRODUCTION

IN RECENT years, micro-small unmanned aerial vehicles (UAVs) have been developed as a result of their outstanding performances in many fields, such as aerial photography, precision farming, and unmanned cargo. An essential type of micro-small UAVs has the capability of vertical takeoff and landing (VTOL), which can hover and take off or land without a runway like helicopters. One of the typical VTOL UAVs, the tail-sitter, tilts its entire body to transform from a hover to a forward flight, which has a simpler tilting mechanism in comparison to tilt-rotor aircraft. However, its attitude control torque is entirely dependent on airspeed over aerodynamic surfaces

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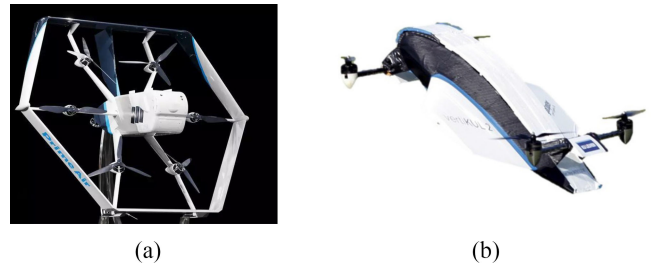


Fig. 1. Examples of a lifting-wing multicopter. (a) Prime air delivery drone [6]. (b) VTOL UAVs with continuously varying transition angle [7].

from a combination of propeller slipstream and incoming flow. It brings along the control difficulty and poor wind resistance. The quadrotor tail-sitter solves these problems by combining a wing with a conventional multicopter. There are some examples of it, such as Quadshot [1] and a quadrotor tail-sitter based on the X5 flying wing [2]. In 2015, a quadrotor tail-sitter applied to parcel delivery was developed [3]. The works about fundamental attitude and position control and optimization of the transition flight are also carried out for a quadrotor tail-sitter [4], [5].

In June 2019, as a type of modified quadrotor tail-sitter with the advantages of higher efficiency and wind resistance, the latest version of its prime air delivery drone [6] was unveiled by Amazon, as shown in Fig. 1(a). A similar UAV configuration is implemented to fly in all flight phases spanning a continuous transition between hover and cruise flight, as shown in Fig. 1(b) [7]. We call these UAVs as *lifting-wing multicopter*, which is a combination of a multicopter with a wing installed at a specific mounting angle less than 90° .

The implement of the lifting-wing multicopter needs an appropriate design method based on an appropriate performance evaluation. There are different design methods for the VTOL UAV. One type of design methods is based on the empirical approach and analysis results [1], [8]–[10]. For the conventional airplane, the sizing method is a representative method based on the empirical approach and analysis results. The main airplane design parameters, i.e., wing loading and thrust loading, can be determined by given means to meet performance requirements [11]. The sizing method is also implemented to VTOL UAV design [12], [13]. This type of method can reveal the relations among the different parameters, which helps designers to make design decisions. On the other hand, this may result in a suboptimal solution. Another method is based on multidisciplinary design optimization (MDO) [14]–[16]. In this method,

the dynamic model, the propulsion model, and the structural model are established first. The global optimal design parameters can be obtained by solving the optimization problem based on the evaluation models. Compared with the analytical design, the optimization process is regarded as a black box and the relations among the parameters that are implicit in the MDO-based method. To let the designer get involved in the optimization process and make the designer's selection and modification easier, it needs to get both the influence of parameter variation and the optimal design. In [17], both the sizing method based on integrated analysis and numerical optimization are used for a fixed-wing VTOL aircraft. An analytical design optimization method, which decouples the optimization design problem into several subproblems, is proposed to design multicopter [18].

As a new kind of VTOL UAVs, the lifting-wing multicopter has not attracted much attention, especially for its performance evaluation and design. It has a specific mounting angle between 0° and 90° , which distinguishes it from any other conventional airplane and VTOL aircraft. Therefore, it brings the problems of propulsion system working in oblique flow and the correlation among some design parameters of the propulsion system and wing caused by a mounting angle, which was not discussed before. Besides, it is the mounting angle that affects the performance largely. There is little research about the evaluation and design method for the VTOL UAV of this type. Motivated by this, a performance evaluation method and a unique concrete design process for the lifting-wing multicopter are proposed in this study. The main contributions of this article are summarized as follows:

- 1) The proposed range and transition time evaluation method takes the mounting angle into consideration. The range expression is rearranged to decompose the dynamic model and the propulsion model for exposing the relation among parameters.
- 2) The concrete sizing optimization method with the consideration of mounting angle is given. The optimal design parameters, including the mounting angle, are found based on the range evaluation and further sensitive study with the constraint of transition time, which can realize the visualization and manual selection of a design point.
- 3) The validity of the proposed optimization design method is verified by the flight experiment.

First, the features of the lifting-wing multicopter are described. Second, the models of each component are established. Then, the range and transition time evaluation methods are obtained. Besides, the evaluation results for conventional VTOL performances are verified. In Section V, the factors affecting the range performance are studied. The concrete design process is proposed in Section VI. Finally, a lifting-wing multicopter prototype was fabricated, and flight experiments were carried out to verify the design results.

II. LIFTING-WING MULTICOPTER DESCRIPTION

The lifting-wing multicopter is a kind of hybrid UAVs [19], which is presented in Fig. 2. It is essentially a multicopter with a blended wing body mounted on its frame. In contrast to the traditional tail-sitter whose wing and propeller are placed at

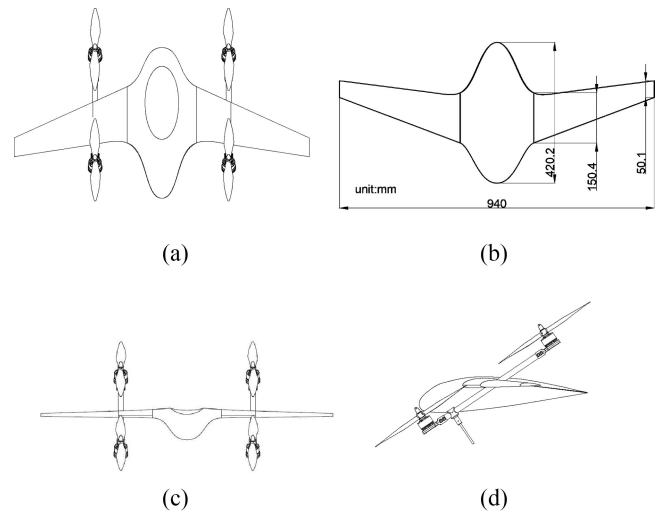


Fig. 2. Conception of the lifting-body multicopter. (a) Top view. (b) Detail of lifting wing. (c) Front view. (d) Side view

about 90° , the lifting-wing multicopter has a specific mounting angle γ , which is less than 90° , as shown in Fig. 5. The wing of the lifting-wing multicopter is a kind of wing-fuselage fusion. It is more like a flattened fuselage, capable not only of accommodating payload but also of generating lift. During the cruise phase, besides the lift from the wing, the propellers also provide force to overcome the gravity, which reduces the wing area. In this phase, the vehicle is controlled via the elevon control surfaces and the differential thrust of propellers together. During the hover phase, the vehicle is completely controlled by propellers.

There are some advantages of lifting-wing design. The wing of lifting-wing multicopter has adequate capacity to carry cargo or other different payloads. Meanwhile, it has a higher lift-to-drag ratio because of the reduced drag from the smaller wetted surface area and the additional lift from the flattened fuselage. Besides, because the attitude of the vehicle is mainly controlled by propellers, there is no need to guarantee stability by using the S-shaped airfoil and empennage. Using the conventional airfoil instead of the S-shaped airfoil further improves the lift-to-drag ratio. The discard of empennage decreases the drag and structure weight. The foregoing feature increases flight efficiency.

This configuration also has better wind resistance and simpler transition. The wing of the lifting-wing multicopter has a smaller reference area and total span, which improves the stability in windy conditions. Besides, the smaller mounting angle between the propellers and the wing decreases the projected area of the wing facing against the wind direction, resulting in the smaller disturbing force in the hover phase. The smaller mounting angle means a smaller transition angle, allowing to fly in all flight phases spanning a continuous transition between hover and cruise flight more simply.

The particular mounting angle gives the vehicle long-range potential, as shown in Fig. 14. Because the specific mounting angle γ is less than 90° , the propellers operating in the oblique flow require less mechanical power compared with operating in axial flow when generating the same force during the cruise

TABLE I
SYSTEM PARAMETERS

Component	Parameters
Status	Pitch Angle $\theta(^{\circ})$, Cruise Velocity $V(\text{m/s})$
Frame	$\Theta_f = \{\text{Gravity } G(\text{Kg}), \text{Rotors Number } n_r, \text{Drag Coefficient } C_D, \text{Lift Coefficient } C_L, \text{Reference Area } S(\text{m}^2), \text{Aspect Ratio } A, \text{Taper Ratio } \lambda, \text{Mounting Angle } \gamma(^{\circ})\}$
Propeller	$\Theta_p = \{\text{Diameter } D_p(\text{inch}), \text{Pitch } H_p(\text{inch})\}$
Motor	$\Theta_m = \{\text{Motor Speed Constant } K_V, \text{No-load Current Coefficient } B_{I0}, \text{Internal Resistance } R_m(\Omega)\}$
Battery	$\Theta_b = \{\text{Capacity } C_b(\text{w} \cdot \text{s}), \text{Voltage } U_b(\text{V})\}$

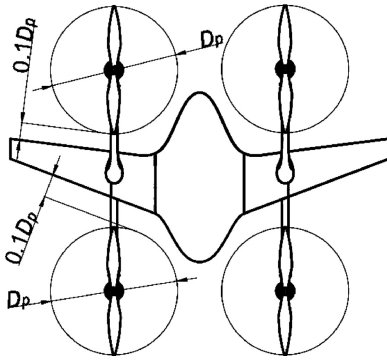


Fig. 3. Location constraint of propellers.

phase [20]. In an oblique flow, the axial component of the freestream airspeed V_0 is equal to the airspeed of the inflow V times the sin value of the pitch angle θ . It turns out that V_0 is smaller than V , resulting in smaller required power of the propeller working in oblique flow. Meanwhile, the smaller axial velocity is more similar to the hover state, which means a better balance between the efficiency of hover and cruise.

III. SYSTEM MODELING

The rapid and accurate modeling for each component is the key to performance evaluation and design. At first, in the specific design, the interaction between propellers and wing turns out to be negligible, so the dynamic model and the propeller model can be considered separately. Then, models for lifting-wing multicopter dynamic, the propeller, the motor, and the electronic speed controller (ESC) are established, respectively. The system parameters are shown in Table I.

A. Avoidance Interaction Between Propellers and Wing

According to [21], the upstream and downstream of the propeller are in a stream tube. There is a well-defined slipstream that separates the flow passing through the propeller disk from that outside the disk. Thus, to avoid the interaction between propellers and wing, it is required to keep the lifting wing outside the slipstream of propellers. The propeller is mounted at the distance of one in ten propeller diameter to the edge of the lifting wing, as shown in Fig. 3.

The propellers–wing interaction should be further studied with given location constraint. A variety of Navier–Stokes-based

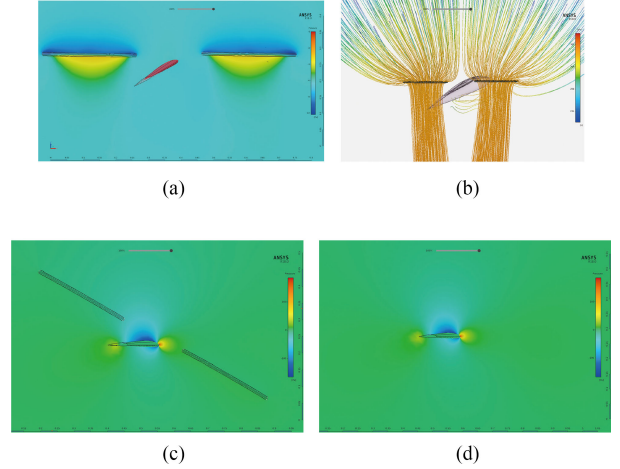


Fig. 4. Interaction between propellers and wing simulation. (a) Pressure distribution in the hover state with propellers. (b) Streamline in the hover state with propellers. (c) Pressure distribution in the cruise state with propellers. (d) Pressure distribution in the cruise state without propellers.

methodologies have been employed for the prediction of the rotor and fuselage interaction [22]. The actuator disk of momentum source methods can be used to analyze the interaction [23]. This method is conducted by ANSYS. For the solver, the pressure–velocity coupling SIMPLEC solver is used.

In the hover state, according to the required thrust and the area of disk, the value of momentum source is set as $3233 \text{ kg} \cdot \text{m}^{-2} \cdot \text{s}^{-2}$. The result is that the force on the lifting wing is 0.045117 N , and there is no significant variations in pressure around its surface shown in Fig. 4(a). Besides, the streamline through the propellers is mainly away from the lifting wing, as shown in Fig. 4(b). It means that the influence of propellers on the lifting wing can be negligible in the hover state.

In the cruise phase, the value of momentum source is set as $530 \text{ kg} \cdot \text{m}^{-2} \cdot \text{s}^{-2}$. The inlet velocity is set to 20 m/s . The result is that the force on the lifting wing with propellers and without propellers is 21.429 and 21.313 N , respectively. In the cruise phase, the force generated on the propellers is much less than the force on the lifting wing, and the total area of propeller disks is larger than the area of the lifting wing. Therefore, the pressure variations caused by propellers are much less than the pressure variations caused by the lifting wing, as shown in Fig. 4(c) and (d). Thus, compared with the result without propellers, the force on the lifting wing and the pressure distribution around its surface does not have much difference with propellers.

Therefore, in both hover and cruise states, the influence of propellers on the lifting wing can be negligible in this distance constraint.

B. Lifting-Wing Multicopter Dynamic Modeling

The dynamic modeling is built for range and transition time performance evaluation. Thus, the analysis of the dynamic of the lifting-wing multicopter is in the longitudinal plane. In addition, it is assumed that the vehicle keeps horizontal linear movement. Therefore, the wind frame expressing the aerodynamic forces on an aircraft is aligned with the Earth frame. The rotation from

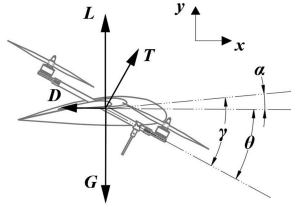


Fig. 5. Longitudinal force analysis of the lifting-wing multicopter.

the body frame to the Earth frame is denoted as

$$\mathbf{R} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}. \quad (1)$$

On the basis of the balance of linear momentum, the translational dynamics of the multicopter is written as [24]

$$\mathbf{F} = m \begin{bmatrix} a_x \\ a_y \end{bmatrix} \quad (2)$$

where m is the mass of aircraft and a_x and a_y are accelerations in x and y directions, respectively. As seen in Fig. 5, the total force $\mathbf{F}$ acting on the lifting-wing multicopter is written as

$$\mathbf{F} = \mathbf{R} \begin{bmatrix} 0 \\ T \end{bmatrix} + \begin{bmatrix} -D \\ L \end{bmatrix} + \begin{bmatrix} 0 \\ -G \end{bmatrix}. \quad (3)$$

For a single propeller, the thrust T_p is expressed as $T_p = T/n_r$, where n_r is the number of rotors, T is the total thrust of multicopter, and G is the gravity of aircraft. The drag D and lift L of the frame are given by

$$L = \frac{1}{2} \rho V^2 S C_L \quad (4)$$

$$D = \frac{1}{2} \rho V^2 S C_D \quad (5)$$

where ρ is the air density. The coefficient of drag C_D and the coefficient of lift C_L are calculated by the vortex lattice method (VLM) and the drag build-up method, respectively, which are described as follows.

The VLM is used to predict the aerodynamic characteristics of the aircraft rapidly. In this method, the wing of the lifting-wing multicopter is modeled with horseshoe vortices distributed along span and chord. The VLM is actualized by the AVL program [25]. However, the defect of the VLM is that the effects of thickness and viscosity are neglected, and only the induced drag coefficient C_{Di} is given, causing the inaccuracy of the total drag. Generally, drag can be divided into two categories: parasite drag C_{Dp} and induced drag C_{Di} . The total drag coefficient is described as

$$C_D = C_{Dp} + C_{Di}. \quad (6)$$

The parasitic drag is also called zero-lift drag comprised of skin friction drag, pressure drag, and interference drag. The parasite drag on different components of an aircraft can be added to the induced drag to get the total drag of the aircraft, which is called the drag build-up method. Using the drag build-up method, the parasite drag coefficient is described as

$$C_{Dp} = \frac{Q}{S} \sum_{i=1}^n K_i C_{fi} S_{weti} \quad (7)$$

where Q is the drag interference factor, S is the reference area, K_i is the form factor, C_{fi} is the skin friction coefficient, and S_{weti} is the total wetted area of the component. Each component of the vehicle is represented by i . Because of the low interference on the whole lifting-wing multicopter, the value of Q is 1.1. The skin friction coefficient is to quantify the friction caused by the viscous effects. The Schlichting formula for estimating the skin friction coefficient on all aircraft components is described as

$$C_f = \frac{0.455}{(\log Re)^{2.58}} \quad (8)$$

where Re is the Reynolds number. The wetted area is the total surface area of a given aircraft component that is exposed in the air.

C. Propeller Modeling

The commercial propellers of one type are identified by two parameters only: diameter D_p and pitch H_p . Thus, the propeller model and design are based on these two parameters. The propeller modeling is based on the vortex theory and blade element theory. In addition, for a multicopter, it is essential to consider that the propellers operate under nonaxial flow conditions. When the lifting-wing multicopter flies at the pitch angle θ , its propellers' angles of inclination relative to the horizontal freestream V is the same as the pitch angle θ . The velocity vertical to the propeller disk V_0 is expressed as

$$V_0 = V \sin \theta. \quad (9)$$

According to [26], the optimal blade number is chosen as 2 for multicopter propellers. Through the vortex theory, the blade can be replaced by a bound vortex system, by which its thrust and torque are calculated without knowing the specific shape of the propeller. At the blade section whose distance from the axis of rotation is r , the strength of the vortex is equal to $\Gamma(r)$, the circulation around the corresponding blade section. It is assumed that the distribution of circulation is ideal, and $\Gamma_{ideal}(r)$ can be expressed. The variable R_p represents the propeller radius, N is the angular velocity, and V_0 is the advance velocity of a two-bladed propeller. Prandtl's tip loss factor F is calculated as

$$F = \frac{2}{\pi} \arccos e^{-f} \quad (10)$$

where f is given by the following expression:

$$f = \left(1 - \frac{\mu}{\mu_0}\right) (1 + \mu_0^2)^{\frac{1}{2}}. \quad (11)$$

The ideal distribution of circulation for the two-bladed propeller is expressed as [27]

$$\frac{\Gamma_{ideal}(r)N}{\pi V_0 W_a} = \frac{\mu^2}{1 + \mu^2} F \quad (12)$$

where $\mu = Nr/V_0$ and $\mu_0 = NR_p/V_0$.

When obtaining the ideal distribution of circulation, the real circulation value is determined by the blade element at $r/R_p = 0.5$, which expressed as

$$\Gamma_{0.5} = \frac{1}{2} W_c C_l \quad (13)$$

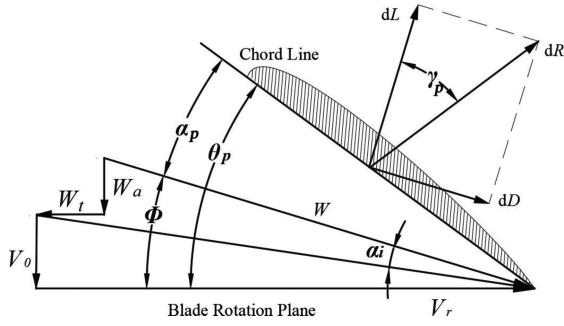


Fig. 6. Blade element at $r/R_p = 0.5$.

where W is the resultant velocity, c is the chord, and C_l is the lift coefficient of the blade element.

Fig. 6 shows a cross section of the blade element at $r/R_p = 0.5$. The parameter V_r is the rotational velocity, equal to Nr . The induced velocity is comprised of two components: the axial W_a and the tangential W_t . The parameters α_i and α_p are the induced and cross-sectional angles of attack, respectively. The geometric pitch angle of the cross section is θ_p , which is related to the parameters of the propeller, as shown in the following expression [28]:

$$\theta_p = \arctan \frac{H_p}{\pi D_p}. \quad (14)$$

The induced velocity in two directions is given by [21, eq. (4.30)], as follows:

$$W_t = \frac{cW C_l(\alpha_p)}{4\pi r F} \quad (15)$$

$$W_a = \frac{1}{2} \left\{ -V_0 + [V_0^2 + 4W_t(V_r - W_t)]^{\frac{1}{2}} \right\}$$

where C_l is a function of the local angle of attack α_p , determined by the geometric relationship shown in Fig. 6. The cross-sectional resultant velocity W is determined by adding the velocity vector. It is shown as

$$W = \left[(V_r - W_t)^2 + (V_0 + W_a)^2 \right]^{\frac{1}{2}}. \quad (16)$$

By solving (15) and (16), the cross-sectional lift coefficient C_l is obtained. Then, the value of circulation $\Gamma_{0.5}$ at $r/R_p = 0.5$ is obtained by (13). The real value of circulation at any radial position is determined by the following expression:

$$\Gamma(r) = \Gamma_{\text{ideal}}(r) \frac{\Gamma_{0.5}}{\Gamma_{\text{ideal}}(0.5R_p)}. \quad (17)$$

Finally, the total thrust T_p and torque M of the two-bladed propeller are obtained by spanwise integration

$$T_p = 2\rho \int_0^{R_p} \Gamma(r) [(V_r - W_t) - \gamma_p (V_0 + W_a)] dr$$

$$M = 2\rho \int_0^{R_p} \Gamma(r) [\gamma_p (V_r - W_t) + (V_0 + W_a)] dr. \quad (18)$$

As shown in Fig. 7, the comparison of computation to experiment results of the APC thin electric 10×7 propeller shows

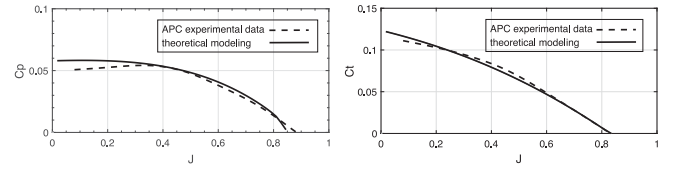


Fig. 7. Propeller model verification.

that the method of propeller performance evaluation is reliable. In the graph, C_p and C_t are the power coefficient and the thrust coefficient, respectively, and J is the advance ratio.

D. Motor Modeling

In this subsection, the equivalent motor current I_m and the equivalent motor voltage U_m are obtained from N , M , and Θ_m . For clarification, the functions to calculate I_m and U_m are expressed as

$$I_m = f_{I_m}(N, M, \Theta_m), \quad U_m = f_{U_m}(N, M, \Theta_m). \quad (19)$$

Motor modeling requires the following assumptions [29].

- 1) The power coefficient is approximately equal to 1 [30].
- 2) Magnetic loss is ignored.

Based on the above assumptions, the input power from the ESC is

$$P_{m_{in}} = U_m I_m. \quad (20)$$

The output power to the propeller is

$$P_{m_{out}} = (I_m - I_{m0})(U_m - I_m R_m) = NM \quad (21)$$

where I_{m0} is the actual motor nominal no-load current and R_m is the internal motor resistance. The motor speed is equal to the propeller speed, which is expressed as

$$N = \frac{2\pi}{60} (U_m - I_m R_m) K_V \quad (22)$$

where K_V is the motor speed constant and $2\pi/60$ is the unit conversion from revolutions per minute (RPM) to angular velocity. The relationship between the internal resistance and the no-load current is established by the statistical data [31]: $I_{m0} = B_{I0}/R_m^{0.6}$, where B_{I0} is the no-load current coefficient; here, $B_{I0} = 0.14$. From the above statement, the functions in (19) are expressed as

$$f_{I_m}(N, M, \Theta_m) = \frac{2\pi}{60} M K_V + \frac{B_{I0}}{R_m^{0.6}}$$

$$f_{U_m}(N, M, \Theta_m) = \frac{60 N}{2\pi K_V} + \frac{2\pi}{60} R_m M K_V + B_{I0} R_m^{0.4}. \quad (23)$$

E. ESC Modeling

Through the ESC modeling, the battery output current I_b is obtained from I_m , U_m , and U_b . For clarification, it is expressed as

$$I_b = f_{I_b}(I_m, U_m, U_b) \quad (24)$$

where U_b is the battery voltage. Assuming that the ESC is lossless, the duty cycle σ is obtained by

$$\sigma = \frac{U_m}{U_b}, \quad \sigma \in [0, 1]. \quad (25)$$

The number of ESCs is equal to that of rotors n_r , so I_b is obtained by

$$I_b = n_r I_e + I_{\text{control}} \quad (26)$$

where I_b is the battery output current and $I_e = \sigma I_m$. I_{control} is the current supplied to the flight controller, and usually, $I_{\text{control}} = 1$ A. Therefore, we have

$$f_{I_b}(\Theta_e, I_m, U_m, U_b) = \frac{U_m}{U_b} I_m n_r + I_{\text{control}}. \quad (27)$$

IV. PERFORMANCE EVALUATION

Compared with the conventional airplane, the lifting-wing multicopter does not have to consider the performances of take-off and landing distance and stall speed. The high thrust-weight ratio of VTOL makes the vehicle satisfy the climbing requirement. Thus, the flight range and the time of transition from hover to forwarding flight are critical performances of the lifting-wing multicopter. For achieving the optimal design, the range and transition time must be quantitatively analyzed. Therefore, in this section, the flight range and the time of transition are evaluated based on the models given above.

A. Range Performance

The range performance evaluated here is the longest flight distance without considering the consumption in hover and transition states. The purpose of range performance evaluation is to construct the objective function for optimal design. Therefore, besides obtaining the range, the form of range evaluation expression needs to contain the important intermediary variable directly to expose the relationship among the design parameters and the range, by which how those different design parameters of the propeller and lifting-wing influence the range can be known. The derivation of the formula of range evaluation is shown as follows.

The direct expression of the range is expressed as

$$R = VT_{\text{fly}} = V \frac{C_b}{P_s} \quad (28)$$

where C_b is the battery capacity in watt seconds and P_s is the power of propulsion system in the cruise state, which is equal to $I_b U_b$. For introducing the propeller and lifting-wing parameters into the range evaluation formula, the efficiency of the propulsion system η_s in the cruise state is given as

$$\eta_s = \frac{TV_0}{P_s} = \frac{TV_0}{I_b(T, V_0)U_b} \quad (29)$$

where the battery output current $I_b(T, V_0)$ is a function of the required thrust T and the axial velocity V_0 , which can be obtained through the propulsion model. According to the lift-drag ratio

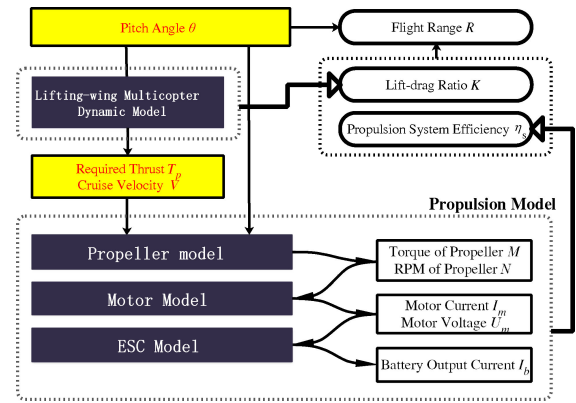


Fig. 8. Range evaluation procedure. The highlighted parameters reflect the influence between the dynamic model and the propulsion model.

$K = L/D$ and (3) in equilibrium, T is obtained as

$$T = \frac{G}{K \sin \theta + \cos \theta} \quad (30)$$

by which the design parameters of pitch angle θ and lift-drag ratio K emerge. Based on (9) and (28)–(30), R is rearranged in form of

$$R = \frac{\eta_s(T, V_0) C_b (K + \cot \theta)}{G} \quad (31)$$

which shows that range R is a function of pitch angle θ and positively correlated with propulsion system efficiency η_s . $\cot(\cdot)$ is the cotangent function equal to $\frac{\cos(\cdot)}{\sin(\cdot)}$. The parameter η_s is a function of the required thrust T , the axial velocity V_0 , and the parameters of the propulsion system, which can be calculated by the propulsion system model. The parameter K can be calculated by the dynamic model, mainly related to the geometry of wing. The range evaluation procedure is shown in Fig. 8.

In the many methods of range evaluation, the parameters are implicit, and relations between each other are also not obvious, which is not conducive to analysis and optimization. In the form of (31), the range evaluation can be decomposed into the propulsion system, pitch angle, aircraft gravity, and lifting-wing aerodynamic. As shown in Fig. 13, the performances of the propulsion system and lifting-wing aerodynamic are represented by two separate parameters, η_s and K , which are conducive to separated analysis for each component. The specific relations, including between η_s and the propulsion parameter, between K and the geometry of the lifting wing, are analyzed in Section V. Thus, the relations among the range the design parameters can be revealed, as shown in Fig. 13. The interaction between the propulsion parameters and lifting wing parameters is reflected by θ , T_p , and V highlighted in Fig. 8. The different design parameters of cruise velocity, pitch angle, and geometry of wing result in different propulsion system efficiency. Thus, the influence is reflected in $\eta_s(T, V_0)$.

In order to verify the accuracy and generality of the proposed range evaluation, the evaluation results are compared with DJI Inspire I, a typical conventional quadcopter, and the quadrotor tail-sitter implemented by the Hong Kong University of Science

TABLE II
DJI INSPIRE I AND QUADROTOR TAIL-SITTER PARAMETERS

Component	DJI INSPIRE I Parameters	Quadrotor Tail-sitter Parameters
Status	$V = 0\text{ m/s}, \theta = 0^\circ$	$V = 12.5\text{ m/s}, \theta = 90^\circ$
Frame	$G = 28.7\text{ N}, n_r = 4, K = 0$	$G = 19.6\text{ N}, n_r = 4, K = 6.7$
Propeller	$D_p = 13\text{ inch}, H_p = 4.5\text{ inch}$	$D_p = 9\text{ inch}, H_p = 6\text{ inch}$
Motor	$K_V = 350, B_{I0} = 0.12, R_m = 0.21\Omega$	$K_V = 980, B_{I0} = 0.21, R_m = 0.13\Omega$
Battery	$C_b = 492480\text{ Ws}, U_b = 24\text{ V}$	$C_b = 259200\text{ Ws}, U_b = 12\text{ V}$

TABLE III
COMPARISON WITH DJI INSPIRE I AND QUADROTOR TAIL-SITTER

Performance	DJI	Proposed	Tail-sitter	Proposed
Power	N/A	417.3W	75.67W	75.1W
Flight endurance	18min	19.7min	56min	57.5min
Flight range	0Km	0Km	42.1Km	43.2Km

and Technology [15]. When the lift-drag ratio K is equal to 0 and the pitch angle θ is equal to 0° , the hover time T_{fly} of a conventional quadcopter in the hover state can be estimated by (28). The range is equal to 0 calculated by (31), which is consistent with the fact. When the pitch angle θ is equal to 90° , the range of the tail-sitter UAV can be obtained. **Table II** shows the detailed parameters of each vehicle. The evaluation results, shown in **Table III**, are compared with the corresponding given data. The results demonstrate that the proposed evaluation method is feasible to estimate the performances of conventional VTOL. It should be noted that the quadrotor tail-sitter will have a large range if the pitch angle θ is smaller. For the given quadrotor tail-sitter, when θ is equal to 40° , the range is 47.6 km, which is larger than the original design. Compared with other evaluation methods, this evaluation method is suitable to a variety of VTOL aircraft such as conventional multicopter, quadcopter tail-sitter, and lifting-wing multicopter because of the consideration of pitch angle θ variation.

B. Time of Transition

Transition time t_{tra} is another important performance of a vehicle. The transition flight strategy of the lifting-wing multicopter is designed for minimum time with constant altitude transition flight. Thus, t_{tra} is the time required for the aircraft to transit from the hover state to the cruise state with keeping altitude constant in the condition of the maximum acceleration. If the transition time is too long, it will not be able to meet the maneuverability requirements. Therefore, the transition time should be a design constraint of the lifting-wing multicopter. For a set of given design parameters, the transition time is evaluated by the simulation of the entire transition flight [32]. The process of transition can be expressed as

$$\dot{v}_{\text{tra}} = a_x \quad (32)$$

$$\text{maximize } a_x(T, \theta_{\text{tra}}) \quad (33)$$

$$\text{subject to } \begin{cases} a_y = 0 \\ N < \frac{N_{\text{max}}}{1.5} \\ 0 < \theta_{\text{tra}} < \gamma \end{cases} \quad (34)$$

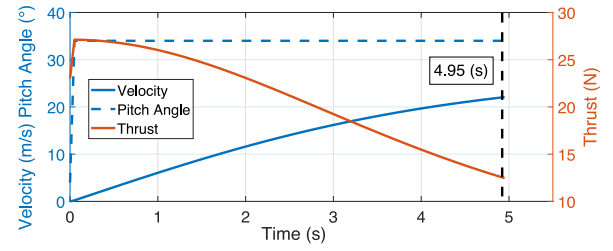


Fig. 9. Results of transition simulation.

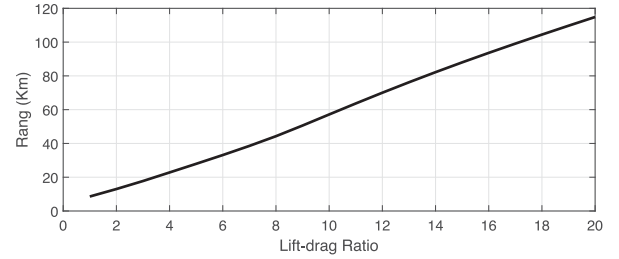


Fig. 10. Range is positively correlated with the lift-drag ratio.

where v_{tra} and θ_{tra} are the flight velocity and the pitch angle as the function of time during the transition, respectively. The value N_{max} is the maximum RPM given by the motor manufacturer. The constant 1.5 is the thrust redundancy for tilting moment generation and disturbance resistance. The value a_x and a_y can be solved by (2)–(5). The each step of the transition process is an optimization problem solved by sequential quadratic programming. The initial transition velocity $v_{\text{tra}}(0)$ is equal to 0. The time when v_{tra} reaches the cruise velocity V is the transition time t_{tra} . The transition time t_{tra} is a function of wing area S , pitch angle θ , and velocity V in the cruise state. When $\theta = 34^\circ$ and $V = 22$ m/s, the transition process is shown in **Fig. 9** and the transition time is 4.95 s.

V. SENSITIVE STUDY

The purpose of the present section is to study the influence of the main parameters of aerodynamics and propulsion systems on the range performance and offer a reference for design. According to (31), the flight range is determined by the efficiency of the propulsion system related to its parameters and the lift-drag ratio related to aerodynamic parameters. Thus, the lift-drag ratio K of the lifting wing, the diameter D_p , pitch H_p , and axial velocity of a propeller on the propulsion system efficiency are analyzed. The effect of the reference area and the aspect ratio of wing on the lift-drag ratio is also studied. For simplicity, the efficiency of the motor is constant. The study includes variations of one or two of the parameters, whereas all other parameters are constant and equal to their nominal values.

A. Lift-Drag Ratio K

According to (29) and (30), the propulsion system efficiency is a function of the lift-drag ratio. According to (31), the range is directly decided by the lift-drag ratio with positive relation. Thus, the relation between η_s and K is plotted in **Fig. 10**. It shows that they are positively correlated.

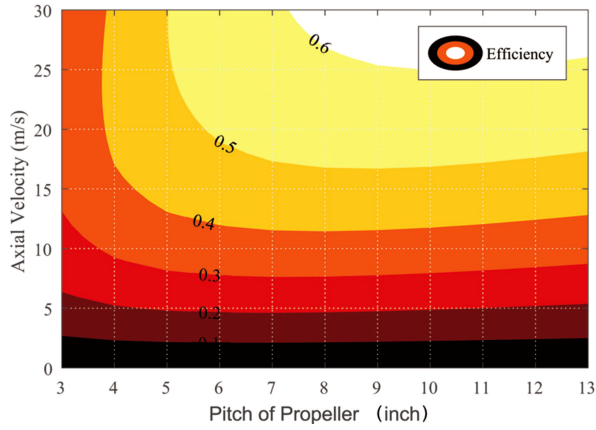


Fig. 11. Efficiency as a function of velocity and pitch of the propeller.

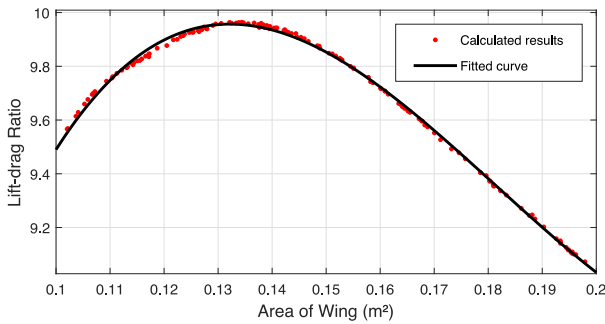


Fig. 12. Lift-drag ratio as a function of the area of the wing.

B. Diameter D_p of the Propeller

Based on the propeller model given above, the propulsion efficiency function of the diameter shows that efficiency increases with the increase in the diameter. It can also be proved by the momentum theory of propeller [21]. With the constant thrust, the longer the diameter is, the smaller the velocity of wake is produced. The smaller velocity of wake means higher efficiency. The diameter should have an upper bound dependent on the geometrical dimensions of the vehicle.

C. Pitch H_p and Axial Velocity V_0 of the Propeller

Because the pitch and axial velocity are coupled in the propeller performance evaluation, the effects of pitch and axial velocity on efficiency are plotted together in Fig. 11. There is an optimal pitch with constant velocity. The efficiency generally increases with the increasing velocity. It can be explained that there is an optimal angle of attack of the propeller blade corresponding to an optimal pitch in the condition of a certain velocity. However, the larger propeller pitch should be limited because a propeller exceeding the pitch upper limit has very low efficiency in the small axial velocity of the hover state.

D. Area S and Aspect Ratio A of the Wing

The wing is a kind of blended wing body with a tapered wing. The area and aspect ratio are the decisive parameters of the lifting body geometry. Based on the VLM program, the lift-drag ratio function of the reference area is plotted in Fig. 12. There

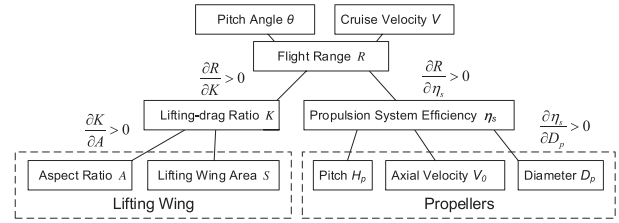


Fig. 13. Relations between the parameters.

is an optimal area with the largest lift-drag ratio. According to the expression of lift, under the condition of constant lift and cruise velocity, the area and the coefficient of lift have one-to-one correspondence and inverse proportional relationship. Through the polar curve, the coefficient of lift with the largest lift-drag ratio can be found. Therefore, the optimal area can be gained by the optimal coefficient of lift. Besides, through the optimal coefficient of lift, the optimal angle of attack is obtained.

According to (31), the aspect ratio is the major factor of the lift-drag ratio. The higher the aspect ratio, the more the lift-drag ratio. However, the exceed high aspect ratio will result in structural problems. The high-aspect-ratio wing is thinner and narrower on its profile. There is less space in the wing for spars. Thus, the aspect ratio should also be limited.

Finally, the relations between the parameters are summarized in Fig. 13.

VI. DESIGN METHOD

In this section, the explicit process of designing the main parameters of the lifting-wing multicopter is proposed. The design method is based on referring to existing conventional UAV parameters, the evaluation models, and the sensitive study in previous sections. The range performance is the design objective, while the time of transition is the design constraint.

A. Step 1: Determination of Dimension and Weight

The VTOL UAV is usually required to deliver packages, perform surveillance, and so on. Depending on the task requirements of the lifting-wing multicopter, the type and size of the payload are given. Referring to the fixed-wing UAV and quadcopter carrying the same payload, the fuselage size and diagonal size of the quadcopter can be basically determined. The diameter of the propeller is designed as long as possible under the limitation of the framework. Then, the weight of the framework can be estimated by referring to a similar UAV. The capacity of the battery is determined based on the reference design and the capacity of the lifting-wing multicopter. Finally, the takeoff weight can be attained by adding up the estimated weight of each component. Besides, for the freight UAV, the maximum capacity is given. Thus, the maximum takeoff weight is attained as a performance requirement.

B. Step 2: Finding Optimal Pitch Angle, Cruise Velocity, and Pitch of the Propeller

The basic parameters have been determined. The lift-drag ratio is estimated as 11. Through the certain pitch angle and

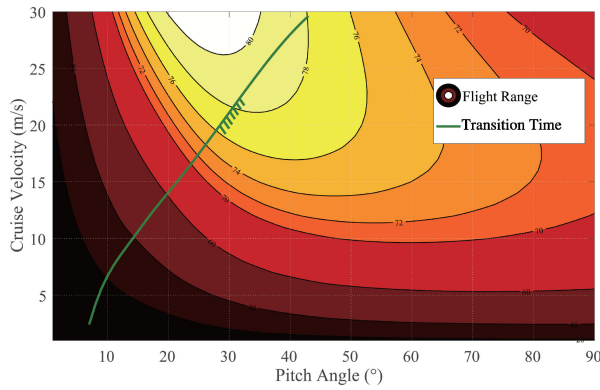


Fig. 14. Range contour map with the boundary of transition time with different wing areas.

cruise velocity, the required thrust T and axial velocity V_0 are obtained by the dynamic model. With the thrust and axial velocity obtained, the highest efficiency of the propulsion system can be obtained through the optimal pitch of the propeller and the optimal motor. The optimal pitch of the propeller is attained with a gradient descent method with the upper bound of 12 in. The optimal efficiency of the motor is constant for simplicity. Thus, the function of range whose variables are pitch angle of frame and cruise velocity with the highest efficiency of propulsion system can be obtained by (31), as shown in Fig. 14. The transition time is the function of cruise velocity, pitch angle, and wing area with a certain weight, aerodynamic parameters, and optimal propeller. It increases with the increase in the cruise velocity and the decrease in the pitch angle. The range contour map is given with the boundary of the transition time with different wing areas in Fig. 14. In the down side of the boundary line, the boundary of the 5-s transition time, the transition time of the design points is less than 5 s, which satisfies the design requirements. The boundary changes with different wing areas. Because the wing area is no more than 0.2 m^2 , the boundary of 0.2 m^2 is the decisive boundary. Thus, through the maximum range point in the down side of the lowest boundary line, the pitch angle and cruise velocity are determined. Finally, the optimal pitch of the propeller can be found with the thrust and axial velocity calculated by the designed pitch angle and cruise velocity.

C. Step 3: Determination of Geometry of the Wing

It needs to consider the taper ratio, reference area, and aspect ratio of the wing for designing its geometry. For the tapered wing, a taper ratio should be designed as 0.4, which is ideal for the weight and drag reduction [33]. Based on the parameters that have been determined, the lift of the wing can be calculated by (3) in equilibrium. Then, through the lift–drag ratio function of the wing reference area, the optimal area and the optimal angle of attack in the cruise state can be obtained. Given the angle of attack and the pitch angle, the mounting angle γ is determined by adding them up. To design a higher aspect ratio for a high lift–drag ratio, the mean aerodynamic chord should be as short as possible. According to the requirements of the place of spars and

TABLE IV
COMPARISON OF DESIGN METHODS

Type	Relation Analysis	Optimization	Constrains	Mounting Angle
analytical Method	•	×	×	×
MDO Method	×	•	Equilibrium, Variables Bound, etc	×
Sizing Method with optimization	•	•	Rate of Climb, Stall Speed, etc	×
Proposed Method	•	•	Transition Time, etc	•

the manufacturing feasibility, the shortest chord can be gained. Therefore, the highest aspect ratio is also determined.

D. Step 4: Determination of Other Parameters

Based on the designed size of propellers and lifting wing, the location of propellers can be determined by the constraint that the distance between the lifting wing and the propeller is larger than one in ten propeller diameter, as shown in Fig. 3, to avoid wing–propeller interaction. Through the comparison of different motor efficiency with the designed propeller, the most efficient motor parameters can be selected by the database provided by manufacturers. The maximum thrust should be over twice the vehicle weight to satisfy the wind resistance requirement. Therefore, the appropriate battery voltage should be chosen to satisfy the maximum thrust. Based on the battery voltage and the maximum current of the selected motor, the appropriate ESC can also be selected.

Remark 1: The description of the design process shows the uniqueness of the method for the lifting-wing multicopter compared with the analytical method [1], [8]–[10], the MDO method [14]–[16], and the sizing method with optimization [17] for VTOL UAV, as shown in Table IV. The proposed design method can find the optimal parameters compared with the empirical and analytical design. This method uses integrated analysis to reveal the effects of the mounting angle with the relation between propulsion and lifting wing, which is implicit in the MDO method. Besides, compared with MDO, this can realize the visualization and manual selection of a design point through Fig. 14 to help designers select the preferred design point.

VII. FLIGHT EXPERIMENT

To demonstrate the validity of the evaluation model and design results, a prototype based on the optimal parameter was built, and its performance data were collected in the flight experiments for comparison purposes.

A. Experimental Prototype

Based on the design method, an experimental lifting-wing multicopter is designed. It refers to DJI Matrice 100. The diameter of the propeller is determined as 12 in, the weight of the battery in 0.3 kg, the takeoff weight is approximately 2.5 kg, and the maximum takeoff weight is 3.5 kg. Then, the optimal pitch angle and the cruise velocity are 32° and 22 m/s, respectively;



Fig. 15. Flight test.

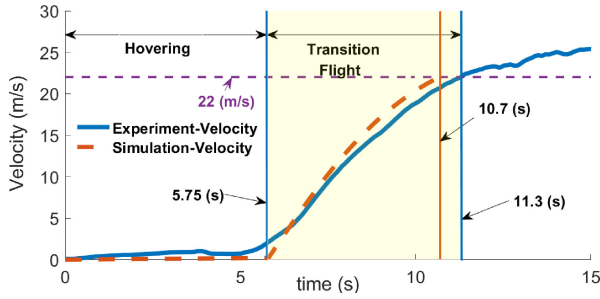


Fig. 16. Experimental results of velocity variation in the transition process.

besides, the pitch angle of the propeller is 12° . The area of the wing is 0.15 m^2 , and the span is 0.94 m . The optimal angle of attack in cruise is 4° . Finally, T-MOTOR MN3510 700-kV brushless motors, 40-A ESCs, and a 4000-mAh 14.8-V LiPo battery were chosen. The wing of the experimental aircraft was cut from a whole piece of EPO foam using a CNC machine. The body frame and spar are carbon fiber. The flight controller is Pixhawk (open source hardware) along with Ardupilot (open source software). We control the lifting-wing multicopter under the multicopter control mode by taking the aerodynamic force and moment as disturbances.

B. Flight Experiment and Result Comparison

As shown in Fig. 15, several flight tests of the lifting-wing multicopter are performed. The takeoff weight in the experiment is 2.1 kg . Through proportional–integral–derivative parameter regulation [34], the aircraft can remain stable from the hover state to the cruise state. When the aircraft entered into equilibrium, the cruise velocity and the pitch angle became steady at 22 m/s and 32° , respectively. The battery output current in this steady state was recorded, and power consumption can be calculated.

The velocity and Euler angles' variations in the process of transition are shown in Figs. 16 and 17, respectively. The simulation of the transition process is described in Section IV-B. Through the velocity variation in Fig. 16, the transition flight begins at $t = 5.75 \text{ s}$ and ends at $t = 11.3 \text{ s}$. Thus, the experimental transition time is 5.55 s , which is 0.6 s longer than the evaluated result. The error is mainly due to the pitch angle response, which is not fast enough compared with the step response in the simulation shown in Fig. 16.

According to the power consumption in the 30-s period, the range performance can be calculated under the condition that

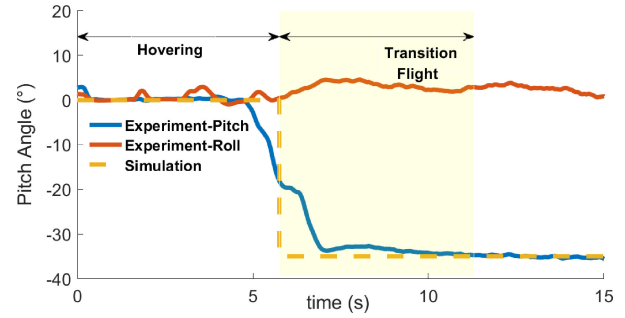


Fig. 17. Experimental results of pitch and roll angle variation in the transition process.

TABLE V
DESIGN AND EXPERIMENT DATA

	Pitch Angle ($^\circ$)	Cruise Velocity (m/s)	Current (A)	Endurance (s)	Range (Km)
Design	32.0	22.0	5.3	340.15	7.48
Experiment	30.5	21.4	6.1	295.08	6.31

the power consumption is 500 mAh. Therefore, the performance values of both the design and the experiment are all obtained, as shown in Table V. The pitch angle, cruise velocity, and current of the experiment are averaged. Through the experiment data, the experimental lifting-wing multicopter is consistent with the design. It indicates the accuracy of the proposed evaluation method.

VIII. CONCLUSION

Many VTOLs have already been designed and implemented, including the quadrotor tail-sitter and the lifting-wing multicopter, which have wind resistance and simple structure that ensure stability. How to predict and extend the range of this kind of VTOL UAV is the key to designing for better performance. In this article, an evaluation method for the range of the lifting-wing multicopter has been proposed based on the models of the dynamic and the propulsion systems. The evaluation method can also apply to other the conventional VTOL UAV. Based on the range expression, the factors affecting the range are analyzed. Then, the practical and concrete design process is given for helping the implementation of this UAV. Through the design process, an experimental prototype is designed for the flight tests, in which the validity of the design and accuracy of the evaluation method are proved.

The evaluation and design method provides compelling support for designing this novel VTOL. This study, therefore, may eventually lead to better performance and wider application of the lifting-wing multicopter. However, the design method using an optimization algorithm needs to be explored in further studies, and to study for the factors affecting wind resistance is also worthwhile.

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